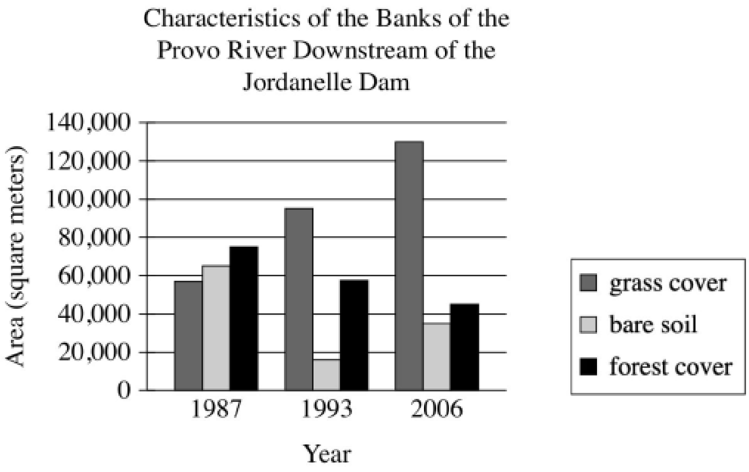
Question 1



The Jordanelle Dam was built on the Provo River in Utah in 1992. Earth scientist Adriana E. Martinez and colleagues tracked changes to the environment on the banks of the river downstream of the dam, including how much grass and forest cover were present. They concluded that the dam changed the flow of the river in ways that benefited grass plants but didn't benefit trees.

Which choice best describes data from the graph that support Martinez and colleagues' conclusion?

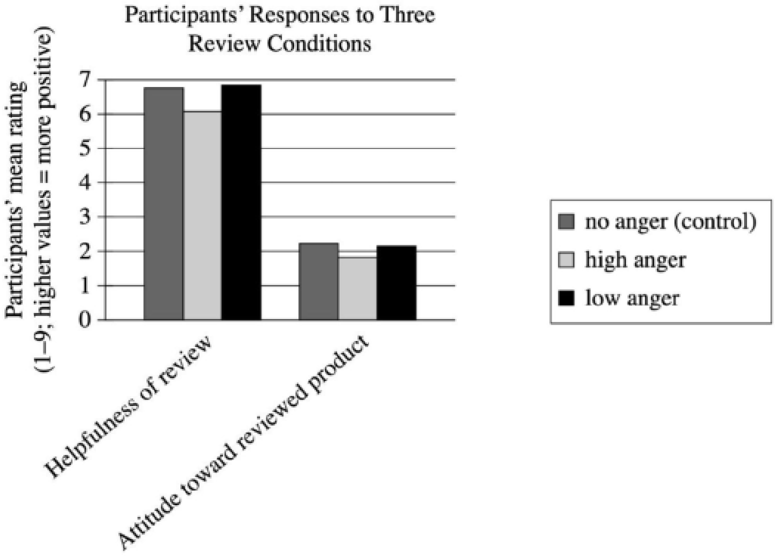
A. The lowest amount of grass cover was approximately 58,000 square meters, and the highest amount of forest cover was approximately 75,000 square meters.

B. There was more grass cover than forest cover in 1987, and this difference increased dramatically in 1993 and again in 2006.

C. There was less grass cover than bare soil in 1987 but more grass cover than bare soil in 1993 and 2006, whereas there was more forest cover than bare soil in all three years.

D. Grass cover increased from 1987 to 1993 and from 1993 to 2006, whereas forest cover decreased in those periods.

Question 2



To understand how expressions of anger in reviews of products affect readers of those reviews, business scholar Dezhi Yin and colleagues measured study participants' responses to three versions of the same negative review—a control review expressing no anger, a review expressing a high degree of anger, and a review expressing a low degree of anger. Reviewing the data, a student concludes that the mere presence of anger in a review may not negatively affect readers' perceptions of the review, but a high degree of anger in a review does worsen readers' perceptions of the review.

Which choice best describes data from the graph that support the students' conclusion?

A. On average, participants' ratings of the helpfulness of the review were substantially higher than were participants' ratings of the reviewed product regardless of which type of review participants had seen.

B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.

C. Participants who saw the low-anger review rated the review as slightly more helpful than participants who saw the control review did, but participants' attitude toward the reviewed product was slightly worse when participants saw the low-anger review than when they saw the no-anger review.

D. Compared with participants who saw the low-anger review, participants who saw the high-anger review rated the review as less helpful and had a less positive attitude toward the reviewed product.

Question 3

Sandra Cisneros's 1984 novella *The House on Mango Street* made a lasting impact on US literature. Its depiction of Mexican American culture inspired later authors to examine their own heritage within their fictional works. Also influential was the book's portrayal of the main character, Esperanza, during a pivotal year of her youth. <u>This insightful depiction of a preteen girl encouraged authors who, like Cisneros herself, are Latina to use fictional works to examine experiences from their own youth.</u>

Which statement, if true, would most strongly support the claim in the underlined sentence?

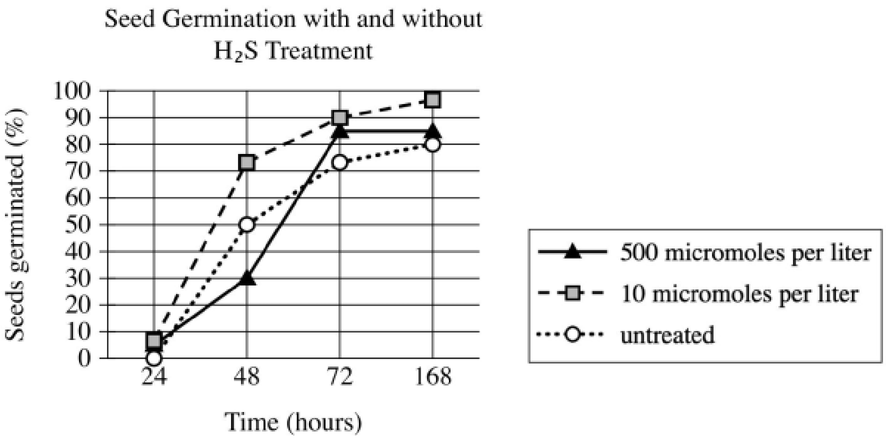
A. In interviews, a number of Latina authors say that *The House on Mango Street* inspired them to write about their own adolescence in their novels.

B. In published writings, several prominent authors who are not Latina say that reading *The House on Mango Street* influenced their approach to writing fiction.

C. The *House on Mango Street* has sold over six million copies and is one of the most commonly read books among high school and university students in the US.

D. Since 1984, new novels about young Latina characters by Latina authors have often been compared to *The House on Mango Street*.

Question 4



In high concentrations, hydrogen sulfide (H_2S) is typically toxic to many plants. Frederick D. Dooley and colleagues wanted to understand what effects low doses of H_2S might have on plant growth. They treated bean, corn, wheat, and pea seeds with various concentrations (measured in micromoles per liter) of H_2S and tracked the germination of those seeds along with the germination of untreated seeds. Treatment with particular concentrations of H_2S was associated with accelerated germination: for example, ______

Which choice most effectively uses data from the graph to complete the statement?

A. at 24 hours, less than 10% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas more than 90% of those seeds had germinated at 168 hours.

B. at 48 hours, more than 70% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 50% of untreated seeds had germinated.

C. at 168 hours, more than 90% of seeds treated with H_2S at concentrations of 10 or 500 micromoles per liter had germinated, whereas less than 70% of untreated seeds had germinated.

D. at 48 hours, approximately 50% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 30% of untreated seeds had germinated.

Question 5

Rivers rich in sediment appear yellow, while increases in red algae make rivers appear red. To track things like the sediment or algae content of large US rivers, John R. Gardner and colleagues used satellite data to determine the dominant visible wavelengths of light measured for various segments of these rivers. The researchers classified wavelengths of 495 nanometers (nm) and below as red, wavelengths between 495 and 560 nm as blue, and wavelengths of 560 nm and above as yellow. The researchers concluded that for the Missouri River, segments flowing into lakes tend to carry more sediment than those flowing out of lakes.

Which finding, if true, would most directly support the researchers' conclusion?

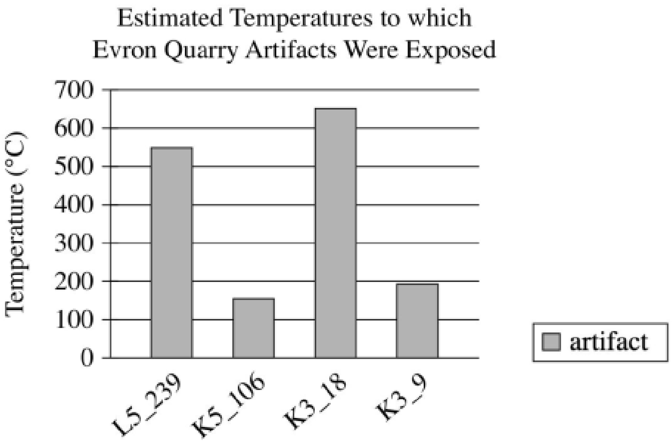
A. The segments of the Missouri River that had higher levels of chlorophyll-a, which contributes to the green color of photosynthetic organisms, have dominant wavelengths of light between 490 and 560 nm.

B. In lakes through which segments of the Missouri River pass, the dominant wavelength of light tended to be above 560 nm near the lakes' shores and below 560 nm in the lakes' centers.

C. The majority of the segments of the Missouri River were found to have dominant wavelengths of light significantly higher than 560 nm.

D. Segments of the Missouri River flowing into lakes typically had dominant wavelengths of light above 560 nm, while segments flowing out of lakes typically had dominant wavelengths below 560 nm.

Question 6



Flint artifacts dating to 800,000 to 1,000,000 years ago have been recovered from the Evron Quarry in Israel. Likely created by the hominin *Homo erectus*, the artifacts have no visual features suggesting that they were exposed to fire, leading some scholars to conclude that these hominins had not acquired control of fire. But Zane Stepka and colleagues recently used a new method to determine whether these artifacts had been exposed to temperatures above 400°C (the typical temperature campfires reach) and concluded that the hominins who inhabited the site may have had control of fire.

Which choice best describes data in the graph that support the team's conclusion?

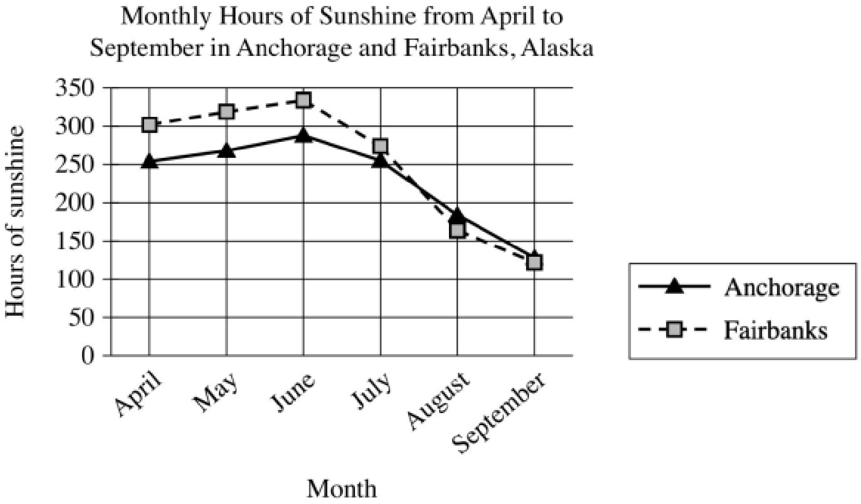
A. Artifacts K5_106 and K3_9 were exposed to temperatures above 400°C.

B. Artifacts L5_239 and K3_18 were exposed to temperatures of approximately 550°C and 650°C, respectively.

C. All of the artifacts were exposed to temperatures above 100°C.

D. Artifact K3_9 was exposed to a higher temperature than was artifact K5_106.

Question 7



A student is researching monthly hours of sunshine in different cities in Alaska. When comparing trends in Anchorage and Fairbanks, the student concludes that the two cities show a similar pattern in the monthly hours of sunshine from April to September.

Which choice best describes data from the graph that support the student's conclusion?

A. The monthly hours of sunshine in both Anchorage and Fairbanks hold steady in June and July before beginning to decline in August.

B. The monthly hours of sunshine in both Anchorage and Fairbanks increase from April to June and then decrease from June to September.

C. Anchorage and Fairbanks both have less than 200 monthly hours of sunshine from April to September.

D. Anchorage and Fairbanks both have more than 300 monthly hours of sunshine from April to June and less than 200 hours from July to September.

Question 8

Average Nitrate and Phosphate Concentrations in Seawater after Volcanic Eruption

Nutrient	Seawater in lava-affected area, 5–45 meters below surface	Seawater in lava-affected area, 75–125 meters below surface	Seawater outside of lava-affected area, 5–45 meters below surface	Seawater outside of lava-affected area, 75–125 meters below surface
Nitrate (micromoles per liter)	3.1	0.4	≤0.03	≤0.01
Phosphate (micromoles per liter)	0.17	0.09	0.14	0.06

After a volcanic eruption spilled lava into North Pacific Ocean waters, a dramatic increase of diatoms (a kind of phytoplankton) near the surface occurred. Scientists assumed the diatoms were thriving on nutrients such as phosphate from the lava, but analysis showed these nutrients weren't present near the surface in forms diatoms can consume. However, there was an abundance of usable nitrate, a nutrient usually found in much deeper water and almost never found in lava. Microbial oceanographer Sonya Dyhrman and colleagues believe that as the lava plunged nearly 300 meters below the surface it dislodged pockets of this nutrient, releasing it to float upward, given that ______

Which choice most effectively uses data from the table to complete the statement?

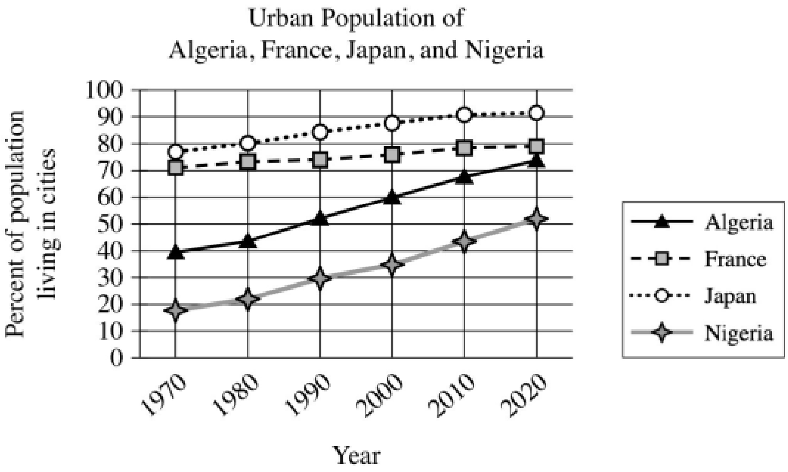
A. at 5–45 meters below the surface, the average concentration of phosphate was about the same in the seawater in the lava-affected area as in the seawater outside of the lava-affected area.

B. for both depth ranges measured, the average concentrations of nitrate were substantially higher in the seawater in the lava-affected area than in the seawater outside of the lava-affected area.

C. for both depth ranges measured in the seawater in the lava-affected area, the average concentrations of nitrate were substantially higher than the average concentrations of phosphate.

D. in the seawater outside of the lava-affected area, there was little change in the average concentration of nitrate from 75–125 meters below the surface to 5–45 meters below the surface.

Question 9



The share of the world's population living in cities has increased dramatically since 1970, but this change has not been uniform. France and Japan, for example, were already heavily urbanized in 1970, with 70% or more of the population living in cities. The main contributors to the world's urbanization since 1970 have been countries like Algeria, whose population went from ______

Which choice most effectively uses data from the graph to complete the assertion?

A. around 50% urban in 1970 to around 90% urban in 2020.

B. less than 40% urban in 1970 to around 90% urban in 2020.

C. less than 20% urban in 1970 to more than 50% urban in 2020.

D. around 40% urban in 1970 to more than 70% urban in 2020.

Question 10

"On Virtue" is a 1766 poem by Phillis Wheatley. Wheatley addresses the poem directly to the quality of virtue, imploring it to assist her in reaching a future goal: ______

Which quotation from "On Virtue" most effectively illustrates the claim?

A. "Attend me, *Virtue*, thro' my youthful years! / O leave me not to the false joys of time! / But guide my steps to endless life and bliss."

B. "I cease to wonder, and no more attempt / Thine height t'explore, or fathom thy profound."

C. "O thou bright jewel in my aim I strive / To comprehend thee. Thine own words declare / Wisdom is higher than a fool can reach."

D. "But, O my soul, sink not into despair, / *Virtue* is near thee, and with gentle hand / Would now embrace thee, hovers o'er thine head."

Question 11

"Odalie" is an 1899 short story by Alice Dunbar-Nelson. In the story, a young woman named Odalie attends the annual Mardi Gras carnival in New Orleans, where she lives with her guardian Tante Louise. Dunbar-Nelson portrays Odalie as eager to escape the monotony of her everyday life: ______

Which quotation from "Odalie" most effectively illustrates the claim?

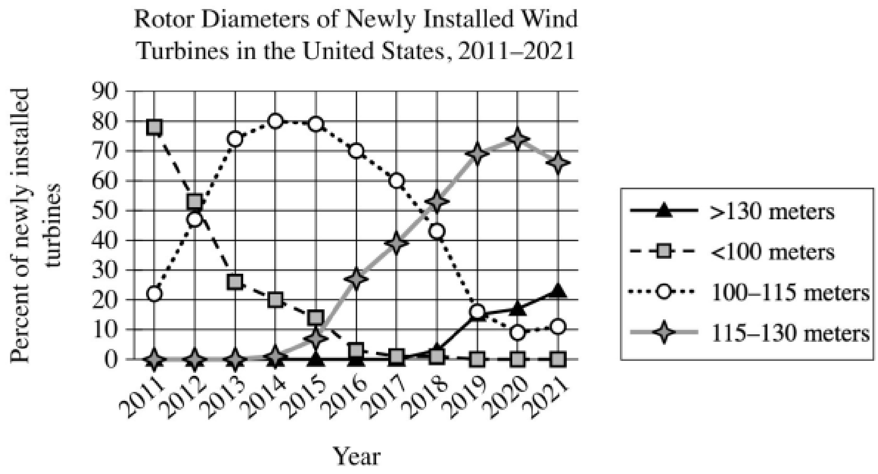
A. "Mardi Gras was a tiresome day, after all, she sighed, and Tante Louise agreed with her for once."

B. "In the old French house on Royal Street, with its quaint windows and Spanish courtyard green and cool, and made musical by the plashing of the fountain and the trill of caged birds, lived Odalie in convent-like seclusion."

C. "When one is shut up in a great French house with a grim sleepy tante and no companions of one's own age, life becomes a dull thing, and one is ready for any new sensation."

D. "It was Mardi Gras day at last, and early through her window Odalie could hear the jingle of folly bells on the [participants'] costumes, the tinkle of music, and the echoing strains of songs."

Question 12



All other things being equal, the larger a wind turbine's rotor diameter (the diameter of the imaginary circle swept by the turbine's rotating blades), the greater amount of energy the turbine can generate. In a research paper on wind power, a student claims that in the United States, the amount of energy generated per newly installed turbine increased substantially between 2011 and 2021.

Which choice best describes data in the graph that support the student's claim?

A. The percentage of newly installed turbines with rotor diameters greater than 130 meters increased every year between 2011 and 2021.

B. In 2011, nearly 80% of turbines installed had rotor diameters of less than 100 meters, whereas only a little more than 20% of turbines installed that year had rotor diameters of 100–115 meters.

C. No turbines installed in 2011 had rotor diameters greater than 115 meters, whereas the majority of turbines installed in 2021 had rotor diameters greater than 130 meters.

D. Most turbines installed in 2011 had rotor diameters of less than 100 meters, whereas most turbines installed in 2021 had rotor diameters of at least 115 meters.

Question 13

Pulitzer Prize–winning writer Héctor Tobar has built a multifaceted career as both a journalist and an author of short stories and novels. In an essay about Tobar's work, a student claims that Tobar blends his areas of expertise by applying journalism techniques to his creation of works of fiction.

Which quotation from a literary critic best supports the student's claim?

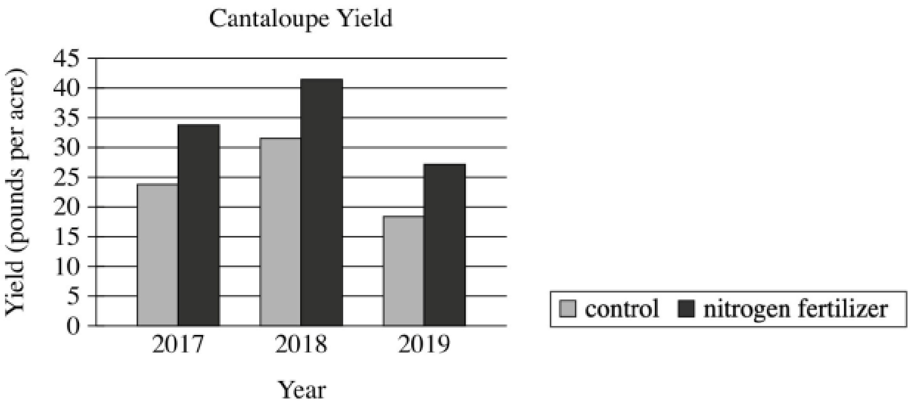
A. "For one novel, an imagined account of a real person's global travels, Tobar approached his subject like a reporter, interviewing people the man had met along the way and researching the man's own writings."

B. "Tobar got his start as a volunteer for *El Tecolote*, a community newspaper in San Francisco, and wrote for newspapers for years before earning a degree in creative writing and starting to publish works of fiction."

C. "Many of Tobar's notable nonfiction articles are marked by the writer's use of techniques usually associated with fiction, such as complex narrative structures and the incorporation of symbolism."

D. "The protagonist of Tobar's third novel is a man who wants to be a novelist and keeps notes about interesting people he encounters so he can use them when developing characters for his stories."

Question 14



To test the effects of a nitrogen fertilizer on cantaloupe production, researchers grew cantaloupe plants and harvested their fruit over three years. In each year, half the plants were grown using a nitrogen fertilizer, and the other half were grown using a control fertilizer that contained no nitrogen. The researchers concluded that the nitrogen fertilizer increases cantaloupe yield.

Which choice best describes data in the graph that support the researchers' conclusion?

A. In every year of the experiment, plants treated with the nitrogen fertilizer had a yield of at least 30 pounds per acre.

B. In every year of the experiment, plants treated with the nitrogen fertilizer had a greater yield than did plants treated with the control fertilizer.

C. The 2018 yield for plants treated with the control fertilizer was greater than was the 2019 yield for plants treated with the nitrogen fertilizer.

D. The yield for plants treated with the nitrogen fertilizer increased from 2017 to 2018.

Question 15

An Ideal Husband is an 1895 play by Oscar Wilde. In the play, which is a satire, Wilde suggests that a character named Lady Gertrude Chiltern is perceived as both extremely virtuous and unforgiving, as is evident when another character says ______

Which quotation from *An Ideal Husband* most effectively illustrates the claim?

A. "Lady Chiltern is a woman of the very highest principles, I am glad to say. I am a little too old now, myself, to trouble about setting a good example, but I always admire people who do."

B. "Do you know, [Lady Chiltern], I don't mind your talking morality a bit. Morality is simply the attitude we adopt towards people whom we personally dislike."

C. "[Lady Chiltern] does not know what weakness or temptation is. I am of clay like other men. She stands apart as good women do—pitiless in her perfection—cold and stern and without mercy."

D. "Lady Chiltern, you are a sensible woman, the most sensible woman in London, the most sensible woman I know."

Question 16

Five of the Responses to Survey about Actions to Conserve Energy

Action	Action category	Percentage of respondents selecting action (%)
Use efficient cars/hybrids	efficiency	2.8
Change thermostat setting	curtailment	6.3
Use bike or public transportation instead of car	curtailment	12.9
Use efficient light bulbs	efficiency	3.6
Turn off lights	curtailment	19.6

In a survey of public perceptions of energy use, researcher Shahzeen Attari and her team asked respondents to name the most effective action ordinary people can take to conserve energy. The team categorized each action as either an efficiency or a curtailment and found that respondents tended to name curtailments more often than they did efficiencies. For example, 19.6% of respondents stated that the most effective way to conserve energy is to turn off the lights, while only ______

Which choice most effectively uses data from the table to complete the text?

A. 6.3% of respondents said it was most effective to use efficient cars or hybrids.

B. 2.8% of respondents said it was most effective to change the thermostat setting.

C. 12.9% of respondents said it was most effective to use a bike or public transportation.

D. 3.6% of respondents said it was most effective to use efficient light bulbs.

Question 17

In 1967 the US Congress created the Corporation for Public Broadcasting, which in turn created National Public Radio (NPR). NPR began producing and distributing high-quality news and cultural programming to affiliate stations across the United States in 1971. In a research paper, a student claims that the Corporation for Public Broadcasting and NPR were inspired by the British Broadcasting System (BBC), which had been established in the 1920s.

Which quotation from a work by a historian would be the most effective evidence for the student to include in support of this claim?

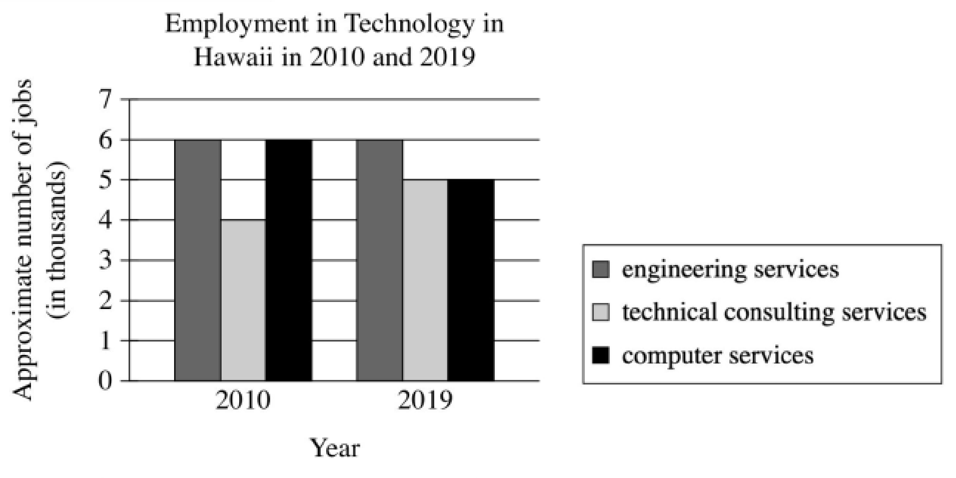
A. "Although the BBC had begun as a private corporation, politicians successfully argued to make it a public company because they believed a public broadcaster could help build national unity in the aftermath of World War I."

B. "For many decades, the BBC had no competition since it held Britain's only broadcasting license, whereas in the United States, the Corporation for Public Broadcasting launched NPR in a broadcasting market already filled with competitors."

C. "Congress's embrace of publicly funded broadcasting reflected a common belief among US politicians that the role of government was not only to ensure people's safety and liberty but also to enrich people's lives in other ways."

D. "The goal of the BBC was to support British democracy by promoting an informed citizenry, and US legislators believed that ensuring access to high-quality programming could do the same for democracy in the United States."

Question 18



A student in Hawaii is interested in pursuing a career in technology and decides to do some research on local trends. The student notices that the number of jobs in computer services in 2010 was ______

Which choice most effectively uses data from the graph to complete the statement?

A. higher than the number of jobs in technical consulting services, and in 2019 was about the same as the number of jobs in engineering services.

B. about the same as the number of jobs in engineering services, and in 2019 was about the same as the number of jobs in technical consulting services.

C. lower than the number of jobs in engineering services, but in 2019 was higher than the number of jobs in engineering services.

D. about the same as the number of jobs in technical consulting services, but in 2019 was lower than the number of jobs in technical consulting services.

Question 19

East Australian humpback whales migrate up to 10,000 kilometers each year to reach their breeding grounds. Researchers long believed that migrating whales live only on the extra energy they stored up during the feeding season. But marine biologist Vanessa Pirotta and her team aren't so sure. They analyzed 20 years of observations of the migrating whales made by citizen scientists (members of the public who help collect data for scientific research). The team claims that the whales may not live only on their stored energy during migration.

Which finding, if true, would most directly support the team's claim?

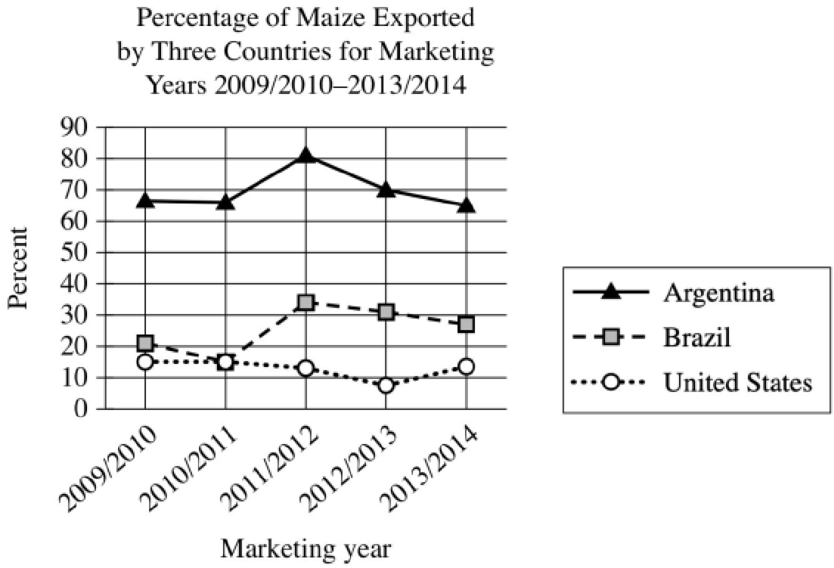
A. Citizen scientists have observed many different types of marine animals feeding alongside the whales.

B. Citizen scientists have made many observations of the whales feeding as they migrate to their breeding grounds.

C. Citizen scientists have made more observations of the whales migrating to their breeding grounds than of the whales returning to their feeding grounds.

D. Citizen scientists have recently begun to observe the whales migrating to their breeding grounds earlier in the year.

Question 20



Argentina, Brazil, and the United States are among the world's leading producers of maize (corn), and each country exports a certain percentage of maize each marketing year, which runs from March to February in Argentina and Brazil and from September to August in the United States. A student is researching those percentages and finds that for the marketing year 2012/2013, the percentage of maize exported by ______

Which choice most effectively uses data from the graph to complete the text?

A. Brazil increased from the previous marketing year but remained lower than the percentage exported by the United States.

B. Brazil exceeded the percentage exported by Argentina for the first time.

C. Argentina decreased from the previous marketing year but remained the highest among the three countries.

D. the United States reached its highest point during the five marketing years.

Question 21

“Often Rebuked, Yet Always Back Returning” is an 1846 poem by Emily Brontë. The poem conveys the speaker’s determination to experience the countryside around her: ______

Which quotation from the poem most effectively illustrates the claim?

A. “Often rebuked, yet always back returning / To those first feelings that were born with me, / And leaving busy chase of wealth and learning / For idle dreams of things which cannot be.”

B. “I’ll walk, but not in old heroic traces, / And not in paths of high morality, / And not among the half-distinguished faces, / The clouded forms of long-past history.”

C. “I’ll walk where my own nature would be leading: / It vexes me to choose another guide: / Where the grey flocks in ferny glens are feeding; / Where the wild wind blows on the mountain side.”

D. “To-day, I will seek not the shadowy region; / Its unsustaining vastness waxes drear; / And visions rising, legion after legion, / Bring the unreal world too strangely near.”

Question 22

Hedda Gabler is an 1890 play by Henrik Ibsen. As a woman in the Victorian era, Hedda, the play’s central character, is unable to freely determine her own future. Instead, she seeks to influence another person’s fate, as is evident when she says to another character, ______

Which quotation from a translation of *Hedda Gabler* most effectively illustrates the claim?

A. “Then what in heaven’s name would you have me do with myself?”

B. “I want for once in my life to have power to mould a human destiny.”

C. “Then I, poor creature, have no sort of power over you?”

D. “Faithful to your principles, now and for ever! Ah, that is how a man should be!”

Question 23

“When Dawn Comes to the City” is a 1922 poem by Claude McKay, who immigrated to the United States from the island nation of Jamaica as an adult. The poem conveys McKay’s contrasting feelings about New York City—his adopted home in the US—and his home country: ______

Which quotation from “When Dawn Comes to the City” most effectively illustrates the claim?

A. “A lonely newsboy hurries by, / Humming a recent ditty; / Red streaks strike through the gray of the sky, / The dawn comes to the city [New York City].”

B. “Dark figures start for work; / I watch them sadly shuffle on, / ’Tis dawn, dawn in New York. / But I would be on the island of the sea, / In the heart of the island of the sea.”

C. “And the shaggy Nannie goat is calling, calling, calling / From her little trampled corner of the long wide lea / That stretches to the waters of the hill-stream falling / Sheer upon the flat rocks joyously!”

D. “The tired cars go grumbling by, / The moaning, groaning cars, / And the old milk carts go rumbling by / Under the same dull stars.”

Question 24

Among the most visited art museums in the world, the Galleria dell'Accademia in Florence had approximately 1.7 million visitors in 2019. The Galleria dell'Accademia also offers virtual tours that art lovers can view online for free. Although there were initial concerns that people who viewed the virtual tours would then consider an in-person visit unnecessary, museum administrators claim that their surveys of in-person visitors show that those concerns were unjustified.

Which statement, if true, would most directly support the administrators' claim?

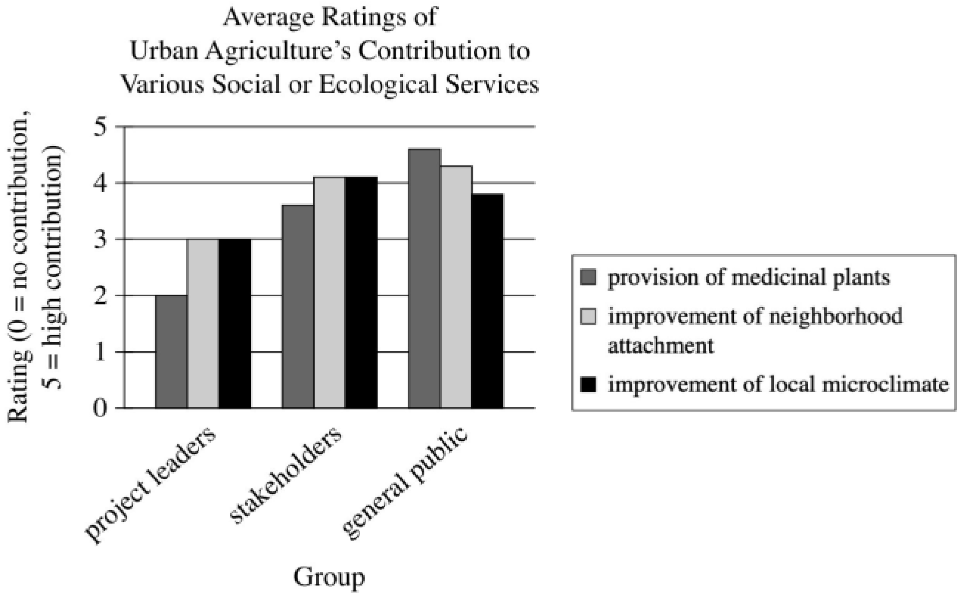
A. Many surveyed visitors to the Galleria dell'Accademia indicated that the virtual tours convinced them to plan an in-person visit.

B. Most surveyed visitors to the Galleria dell'Accademia indicated that they were unaware of the virtual tours before their first in-person visit.

C. Most surveyed visitors to the Galleria dell'Accademia indicated that they lived somewhere other than Florence.

D. Many surveyed visitors to the Galleria dell'Accademia indicated that they would likely view the virtual tours in order to reminisce about their in-person visit.

Question 25



Urban agriculture is the practice of growing plant- or animal-based products in urban settings such as community gardens and rooftop farms. Esther Sanyé-Mengual, Kathrin Specht, and their team surveyed three groups of people in Bologna, Italy—leaders of urban agriculture projects, stakeholders in urban agriculture (e.g., food researchers and urban farming associations), and the general public—to compare their views about the extent to which urban agriculture contributes to 25 social or ecological services that the team identified. The survey results show that, on average, project leaders rated urban agriculture as contributing less to ______

Which choice most effectively uses data from the graph to complete the assertion?

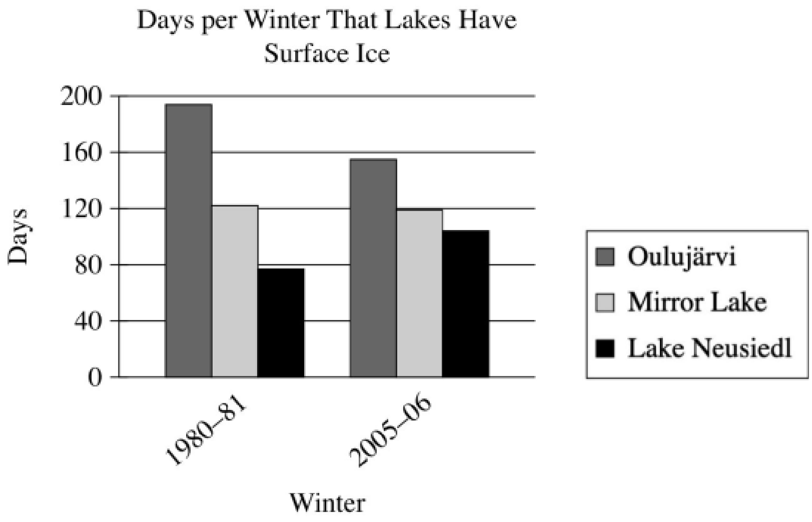
A. providing medicinal plants than to improving the local microclimate.

B. improving community members' sense of attachment to the neighborhood than to offering educational opportunities.

C. improving the local microclimate than to providing medicinal plants.

D. improving the local microclimate than to providing tourist attractions.

Question 26



It is common for freshwater lakes near or above a latitude of 45° north of the equator, like Lake Mjøsa in Norway, to accumulate surface ice in winter. The amount and duration of ice depends on many factors, including local weather conditions as well as the lake's depth, volume, and surface area, but a climate researcher claims that some lakes in these latitudes have seen a decline in the duration of ice between the early 1980s and the mid-2000s. She cites as a typical example ______

Which choice most effectively uses data from the graph to complete the researcher's example?

A. both Lake Neusiedl and Oulujärvi, which had fewer than 195 days of ice in the winter of 1980–81.

B. Lake Neusiedl, which had more days of ice in the winter of 2005–06 than it did in the winter of 1980–81.

C. Oulujärvi, which had fewer days of ice in the winter of 2005–06 than it did in the winter of 1980–81.

D. both Lake Neusiedl and Oulujärvi, which had more than 105 days of ice in the winter of 2005–06.

Question 27

Survey Results for Two Online Account Sign-in Methods

Sign-in method	Percent of participants in the UK who chose method	Percent of participants in Japan who chose method	Percent of participants in India who chose method
Biometrics (for example, a face scan)	33	29	22
Onetime passcodes	16	8	25

A survey listed methods for signing into online accounts. Participants in different countries were asked to choose the sign-in method they view as most secure. The table presents data for two of the methods. According to the table, onetime passcodes were viewed as most secure by ______

Which choice most effectively uses data from the table to complete the text?

A. 33 percent of survey participants in the UK.

B. 22 percent of survey participants in India.

C. 8 percent of survey participants in Japan.

D. 16 percent of survey participants in India.

Question 28

North American Thrasher Mean Bill Size and Habitat Temperature Range

Species	Mean bill surface area (cm^2)	Mean maximum temperature of warmest month (°C)	Mean minimum temperature of coldest month (°C)
Brown thrasher	1.86	30.40	−4.29
Bendire's thrasher	1.98	36.57	0.24
Long-billed thrasher	2.24	35.27	8.82
Cozumel thrasher	2.28	33.27	18.21
Ocellated thrasher	3.26	27.56	5.45

It has been hypothesized that since birds can dissipate excess heat through their bills, bill size should increase with habitat temperature. To evaluate this hypothesis for a 2021 study, Charlotte Probst and colleagues gathered data on mean bill surface area of species of North American thrashers (genus: *Toxostoma*) as well as on climate conditions of the birds' native habitats. Based on their data, Probst and colleagues concluded that the hypothesis was not fully supported.

Which choice most effectively uses data from the table to support Probst and colleagues' conclusion?

A. Although the Bendire's thrasher has one of the smallest mean bill surface areas of the birds included in the table, its habitat has one of the lowest mean maximum temperatures in the warmest month and one of the lowest mean minimum temperatures in the coldest month.

B. Although the Cozumel thrasher has the second greatest mean bill surface area of the birds included in the table, its habitat's mean temperature in the warmest month is significantly higher than that of the other birds' habitats.

C. Of the birds included in the table, the brown thrasher has the smallest mean bill surface area and the habitat with the lowest mean minimum temperature in the coldest month, while the long-billed thrasher has the second-largest mean bill surface area and the habitat with the second-highest mean minimum temperature in the coldest month.

D. Of the birds included in the table, the ocellated thrasher has the largest mean bill surface area and the habitat with the lowest mean maximum temperature in the warmest month, while the Bendire's thrasher has the second smallest mean bill surface area and the habitat with the highest mean maximum temperature in the warmest month.

Question 29

Baltimore, Maryland, has installed engineered structures along 71% of its shoreline to protect infrastructure from wave erosion and other hazards, a practice known as shoreline hardening. To evaluate the responses of waterbirds to two types of hardening structures—riprap and bulkheads—Diann Prosser et al. surveyed waterbird communities consisting of the tundra swan, the great blue heron, and 62 other species at different sites in the Chesapeake Bay on the US East Coast. Utilizing the Index of Waterbird Community Integrity (IWCI), on which a high score corresponds to high community integrity, the researchers found that bulkheads are more strongly negatively correlated with waterbird community integrity than is riprap.

Which finding, if true, would most directly illustrate the researchers' finding?

A. The difference in average IWCI scores for waterbird communities at Stony and Old Road, two sites with a higher percentage of shoreline consisting of bulkheads than of riprap, was statistically insignificant.

B. Waterbird communities at Old Road, a site with a relatively high percentage of shoreline consisting of bulkheads, had lower average IWCI scores than did waterbird communities at Miles, a site with a relatively high percentage of shoreline consisting of riprap.

C. Waterbird communities at Curtis, a site with a high percentage of shoreline consisting of bulkheads and riprap, had lower average IWCI scores than did waterbird communities at Onancock, a site with a low percentage of shoreline consisting of bulkheads and riprap.

D. Waterbird communities at Curtis, a site with equal percentages of shoreline consisting of bulkheads and riprap, had higher average IWCI scores than did waterbird communities at Miles, a site with different percentages of shoreline consisting of bulkheads and riprap.